



**Agentic & Autonomous
Business Intelligence Platform -
PRD**

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1. Executive Summary & Vision

1.1 Vision Statement

To build an autonomous, agent-driven Business Intelligence product that not only visualizes business performance but actively analyzes data, generates insights, identifies business events, determines appropriate actions, and assigns ownership to relevant stakeholders without requiring manual intervention.

The platform will operate through a single unified dashboard with multiple views and will function as a standalone product without dependency on third-party BI tools.

1.2 Problem Statement & Market Opportunity

Today's BI platforms stop at dashboards. Business teams are left to manually:

- Cleanse, transform, and model data before it's usable
- Define, maintain, and recalculate KPIs as business context changes
- Read through reports to detect anomalies or emerging trends
- Decide who should act on an insight and follow up on execution

This creates decision lag. Data arrives; action is delayed. Organizations lose revenue opportunities and suffer avoidable operational failures. The autonomous BI market is rapidly expanding as AI capabilities mature — purpose-built agentic platforms that close the loop from data to action represent the next category.

1.3 Mission

Build a production-grade, multi-agent BI product that ingests business data from any source, auto-generates certified KPIs and dashboards, detects opportunities and risks, autonomously assigns corrective tasks, and continuously learns from user feedback.

1.4 Strategic Goals & Business Objectives

Goal	Objective	Success Indicator
Universal Data Access	Connect any structured database or API with plugin connectors	5+ connector types live at launch
Autonomous Insight	Detect anomalies and generate categorised insights without human trigger	< 5-min insight latency
Governed KPIs	AI-generated KPIs with HITL approval and full audit trail	100% certified KPIs before dashboard activation
Close the Action Loop	AI assigns and executes corrective actions across channels	Task assignment accuracy > 90%
Explainability by Default	Every insight traceable to source data and calculation	100% insights with confidence score + rationale
Standalone Product	No third-party BI tool dependency	Zero Tableau/PowerBI calls in production

2. Target Market & User Personas

2.1. Target Market

Mid-to-large enterprises with fragmented data across multiple sources and operational teams who need faster, more reliable insight-to-action cycles. Initial focus: retail, e-commerce, SaaS, and supply chain businesses with structured transactional data.

User Personas

Persona	Role & Context	Core Need
Executive Sponsor	C-suite or VP. Reads AI summaries and strategic KPIs. Approves high-stakes action plans.	Executive dashboard view. Strategic insights. Approval inbox. High-level reports.
Business Manager	Dept. head managing OKRs and team performance. Receives task assignments and insight alerts.	Manager dashboard. Department insights. Task inbox. Drill-down on KPIs.
Data / BI Analyst	Connects sources, governs KPIs, certifies dashboards, reviews agent outputs.	Full data layer access. KPI builder. Dashboard studio. Agent output logs.
Operations User	Receives and executes tasks assigned by the platform. Reports on resolution.	Task inbox only. Relevant operational KPIs. Insight cards linked to their tasks.

3. Product Scope & Core Modules

Single unified web product with one dashboard and multiple contextual views per role. All intelligence generated in-house via the agent framework. No embedded third-party BI tools.

Module	Description	MVP
Universal Data Connectors	Plugin-based connectors for databases, APIs, flat files	Yes
Dataset Management	Schema detection, sync config, data quality scoring	Yes
KPI Engine + HITL	AI KPI generation, approval workflow, certification, governance	Yes
Dashboard Builder	No-code drag-and-drop builder, widget marketplace, versioning	Yes
Multi-Level Insight Engine	5 insight categories, time-based intelligence, role-filtered delivery	Yes
Explainability Layer	Source tracing, calculation visibility, confidence scores, rationale	Yes
Autonomous Action Engine	Insight-to-action pipeline, multi-channel execution, approval flow	Yes
Feedback Loop	Accept/reject insights, agent learning over time	Yes
Audit & Compliance Center	Full activity logs, AI decision history, governance reporting	Yes
Report Generator	AI-written summaries, scheduled + event-triggered	Yes
Conversational BI Copilot	NL query interface on any dashboard view	Phase 2
Simulation & What-If Analysis	Scenario planning, metric impact maps, forecasting	Phase 2

4. Key Features & Functional Requirements

4.1 Data Connectivity & Dataset Management

Universal Data Connectors (Plugin Architecture)

Each connector is a self-contained plugin registered in the Connector Registry. New connectors can be added without changes to the core platform.

- Relational databases: PostgreSQL, MySQL, SQL Server, Snowflake
- NoSQL: MongoDB
- APIs: REST and GraphQL endpoints with configurable auth (API key, OAuth, Bearer token)
- Flat files: CSV, JSON upload with auto-schema detection
- Future plugin slots: Salesforce, HubSpot, Google Analytics, custom webhooks

Dataset Management

- Schema auto-detection on first connection; field type inference and naming suggestions
- Configurable sync frequency per connector (real-time, hourly, daily, on-demand)
- Data quality scorecard per dataset: completeness, consistency, recency, null rate
- Low-quality datasets flagged and quarantined before entering the agent pipeline
- Incremental sync with full-refresh fallback on schema change detection
- Connector audit log: every sync event, row count, error, and retry recorded

4.2 KPI Management & Governance

AI-Assisted KPI Builder

- LLM analyzes connected schema — field names, types, relationships — to auto-recommend KPIs
- KPI hierarchy creation: company level → department → function → individual metric
- KPI dependency mapping: identifies which KPIs are computed from shared base metrics
- Business glossary integration: KPI names and definitions aligned to org-defined terminology
- AI suggests chart type, benchmark range, and alert threshold per KPI on creation

Human-in-the-Loop (HITL) Governance

- Every AI-generated KPI enters a Pending Certification state — not published until approved
- KPI approval workflow: Analyst proposes → Business Owner reviews → Finance/Leadership signs off
- Certification badge displayed on all certified KPIs visible to end users
- KPI change approval process: any formula or definition edit requires re-certification
- Business owner assignment: each KPI has a named owner accountable for its accuracy

KPI Governance Controls

- Full version control: every KPI definition change stored with author, timestamp, and reason
- KPI audit trail: complete history of approvals, edits, and ownership changes
- Certified vs. Draft KPI distinction enforced across all dashboard views and insights
- KPI health monitoring: stale feeds, formula errors, or low data volume flagged automatically

4.3 Dashboard Builder & Visualization Studio

No-Code Dashboard Creation

- Drag-and-drop canvas: add, resize, and reorder widgets without writing code

- Widget marketplace: KPI tile, line chart, bar chart, heatmap, table, text block, metric card
- Layout customization: grid-based layout with snap-to-grid and responsive breakpoints
- Responsive design: dashboards render correctly on desktop and tablet

Custom Chart Builder

- Analyst selects dimensions, measures, and chart type from a visual configurator
- AI chart recommendation: system suggests optimal visualization based on data shape
- Automatic axis labelling, colour theming, and period comparison overlays

Dashboard Governance

- Dashboard versioning: users save named versions and restore any prior state
- Dashboard types: Personal (private), Team (shared within a group), Executive (org-wide read-only)
- Dashboard templates: pre-built layouts for Sales, Finance, Operations, and Marketing
- Dashboard sharing: share a view with specific users or roles with view-only or edit permissions
- Dashboard approval workflow: Team and Executive dashboards require Analyst or Admin sign-off before publish

4.4 Multi-Level AI Insights Engine

The system continuously monitors all certified KPIs and generates categorised insights. Insight types listed below are representative — actual insights generated are entirely driven by the connected data, certified KPIs, and configured thresholds.

Insight Categories

Category	Example Insight Types	Triggered By
Revenue Insights	Growth/decline, product performance, market shifts	Revenue, sales, order KPIs
Operational Insights	Process bottlenecks, SLA breaches, resource utilisation	Ops, fulfilment, capacity KPIs
Customer Insights	Churn trends, satisfaction drops, engagement patterns	CRM, support, NPS KPIs
Strategic Insights	Competitor movements, market opportunities, growth risks	Data signals + internal KPIs
Financial Insights	Margin compression, cost trend anomalies, profitability drivers	P&L, cost centre KPIs

Time-Based Intelligence

- Month-over-Month (MoM), Quarter-over-Quarter (QoQ), Year-over-Year (YoY) comparisons built into every insight
- Trend detection: slope analysis over configurable 7/30/90-day look-back windows
- Seasonality detection: baseline adjusted for known seasonal patterns to reduce false positives
- All time-period comparisons use certified KPI snapshots only — no draft data in insights

4.5 Role-Based Intelligence Layer

Insights are filtered, ranked, and presented differently per role. The same underlying anomaly generates different insight cards depending on the user viewing it.

Role	Insight Focus	Delivery Format
Executive	Strategic and financial insights only. Macro trend. Business impact summary.	High-level narrative card. No raw data. Approvals queue for P1 actions.
Business Manager	Department-scoped insights. Operational and revenue KPIs for their team. Task-linked insights.	Insight feed with priority flag. Linked to task inbox. Drill-down enabled.
Data / BI Analyst	All insights including low-confidence ones. Data quality flags. Agent decision rationale.	Full insight detail with source trace, confidence score, and feedback controls.
Operations User	Only insights linked to tasks assigned to them. Operational context only.	Task card with embedded insight context. No standalone insight feed.

4.6 Explainability & Trust Layer

Every AI-generated insight includes a full explanation trail. Users can inspect any insight at any level of depth.

- Source data visibility: which dataset, which fields, which time window fed this insight
- KPI calculation visibility: formula used, certification status, version number
- Dataset traceability: connector name, last sync time, data quality score at time of insight
- Insight generation rationale: plain-English explanation of why the anomaly was flagged
- Supporting evidence display: the underlying data points that triggered the detection
- Confidence score (0-100): displayed on every insight card; low-confidence insights visually distinguished
- Data freshness indicator: timestamp of the most recent data that contributed to the insight

4.7 Autonomous Actions & Workflow Execution

Action Flow

Insight Detected → Recommended Action Created → Approval Requested → Human Approval → Action Executed → Notification Sent

Action Channels

- Email: stakeholder notification with insight summary, recommended action, and approval link
- Jira ticket creation: structured ticket with title, description, priority, and assignee from KPI context
- Slack / Teams notification: real-time message to configured channel with insight card and action button
- CRM task creation: task created in connected CRM (Salesforce, HubSpot) with full context
- Internal workflow assignment: task created in the platform's own task management system

Approval Controls

- P1 (critical) actions require explicit human approval before execution
- P2 actions auto-execute after configurable delay unless rejected
- P3 informational actions execute immediately with notification only
- All action decisions logged with approver identity, timestamp, and outcome

4.8 Insight & KPI Feedback Loop

- On every insight card: Accept, Reject, or Provide Feedback controls
- Rejected insights suppressed from the same user's feed for the same KPI pattern for a configurable period
- Feedback data stored as labelled training signals — used to tune anomaly sensitivity per KPI and per role
- KPI feedback: users flag KPI definitions that produce misleading insights; routed back to the HITL approval queue
- Agent learning: feedback signals reviewed weekly by the Insight Agent to adjust detection thresholds; no fully autonomous model retraining in MVP
- Tenant-Specific Agent Tuning (Localized Learning): All feedback signals (Accept/Reject/Provide Feedback) and agent learning are strictly isolated to the specific Enterprise environment. Optional: Only anonymized, non-proprietary structural improvements (e.g., "how to better format a table") can be moved to a Global Model.

4.9 Audit & Compliance Center

- Complete activity log: every user action, data sync, agent decision, and system event with timestamp and user ID
- AI decision history: full record of every insight raised, action recommended, and agent reasoning chain
- Approval history: KPI certifications, dashboard approvals, and action approvals with approver identity
- Governance reporting: exportable compliance report covering KPI certification status, user access changes, and data source activity
- Retention policy: configurable log retention (default 12 months); immutable once written

4.10 Agent Framework

Specialized agents operate in a coordinated, event-driven pipeline. Each agent has a defined input event, output schema, and confidence score.

Agent	Responsibility
Data Agent	Monitors connector health, triggers sync, runs quality validation, routes clean datasets downstream.
KPI Agent	Analyzes schema to generate KPI recommendations. Monitors certified KPIs for drift. Emits kpi_updated events.
HITL Workflow Agent	Manages KPI and dashboard approval queues. Routes items to correct approvers. Tracks SLA on pending approvals.
Reporting Agent	Composes AI narrative summaries per role. Schedules and delivers reports. Triggered by schedule or insight resolution.
Insight Agent	Runs anomaly detection and trend analysis on certified KPI streams. Assigns category, severity, and confidence to each insight.
Explainability Agent	Constructs the evidence trail for every insight: source trace, calculation audit, rationale generation.
Decision Agent	Evaluates insight severity and business impact. Determines action type, priority, and approval requirement.
Task Assignment Agent	Maps required action to correct stakeholder. Creates task records. Monitors overdue items and escalates per SLA.

4.11 Phase 2: Simulation & What-If Analysis

Scoped for Phase 2 delivery.

- Scenario planning: revenue forecasting, cost optimization simulations, resource planning
- What-If engine: user inputs a variable change (e.g., marketing spend +10%, churn +5%) and the simulation engine projects KPI impact
- Metric Impact Maps: visual representation of KPI dependencies, cause-and-effect chains, and business impact across the hierarchy
- Simulation results stored as named scenarios; multiple scenarios comparable side by side

4.12 Phase 2: Conversational BI Copilot

Embedded chat interface on any dashboard view. Lower implementation priority; delivered after core insight and action pipeline is stable.

- Natural language queries: 'Why did revenue drop?', 'Show top-performing regions', 'Compare this quarter vs last quarter'
- Copilot draws on certified KPIs, active insight feed, and RAG context to answer
- Responses include source citations (which KPI, which dataset, which time period)

4.13 Phase 2: Multi-Tenant Admin Console

- Data Isolation Settings: Admin toggle to choose between "Shared Cluster" (Logical Isolation via Row-Level Security) or "Dedicated Instance" (Physical Isolation for high-compliance domains).
- BYOK Management: Interface for Enterprise Admins to provide and rotate their own encryption keys (AWS KMS or GCP KMS integration).
- Audit Log Export: Ability to push tenant-specific audit logs (from Section 4.9) into the client's own SIEM (e.g., Splunk, Datadog).

4.12 Phase 3: Collaborative Intelligence & Annotations

Scoped for Phase 2 delivery to enable team discussion and collective investigation before an insight is converted into a formal task.

- In-App Threaded Comments: Every "Insight Card" and "KPI Tile" includes a comment sidebar allowing for real-time discussion.
- Contextual @Mentions: Users can mention specific team members or roles (e.g., @Finance_Team). Mentions trigger in-app notifications and external alerts via configured channels (Slack/Email).
- Data-Point Annotations: Ability to "pin" a comment to a specific coordinate on a chart or a specific row in a table. This annotation remains visible to all users with access to that view.
- Insight-to-Draft Flow: Comments serve as the "pre-action" stage. Discussion logs can be optionally attached to the final Task/Jira ticket for historical context.
- Agent Participation: The "Reporting Agent" can be @mentioned to ask for a summary of the discussion or to pull additional data into the thread.

4.13 Phase 3: Secure External Sharing & Guest Views

Scoped for Phase 2 delivery to allow enterprises to securely share performance data with external stakeholders (vendors, investors, or board members) without requiring a full user license.

- Public/Guest Link Generation: Authorized admins can generate secure, obfuscated URLs for specific dashboards or reports.
 - Granular Access Control:
 - Read-Only Access: Guests cannot edit dashboards, leave comments, or trigger actions.

- Restricted Drill-down: Toggles to disable "Deep Dive" or "Source Data" views for external links to protect underlying raw data.
- Filtered Views: Ability to share a "filtered version" of a dashboard (e.g., sharing a vendor-specific dashboard where they only see their own performance data).
- Sharing Security:
 - Password Protection: Mandatory or optional password requirements for external links.
 - Expiration Dates: Links automatically deactivate after a set period (e.g., 24 hours, 7 days).
 - Domain Whitelisting: Restrict link access to specific email domains (e.g., only @investor-firm.com).
- Guest Analytics: Audit logs (Section 4.9) track when external links are accessed, by which IP, and for how long.

5. User Journey

End-to-end product experience from initial setup through autonomous insight-to-action, incorporating all platform layers.

Step 1 - Onboarding & Data Connection

- Admin selects a connector plugin (e.g., PostgreSQL) from the marketplace and configures credentials
- Data Agent validates connection, detects schema, profiles fields, and produces a data quality scorecard
- Schema fingerprint stored in the Connector Registry; sync schedule configured

Step 2 - KPI Generation & Certification

- KPI Agent analyzes schema and emits a set of draft KPI recommendations with definitions, formulas, and suggested owners
- HITL Workflow Agent opens an approval queue: Analyst reviews → Business Owner approves → Finance signs off
- Certified KPIs published to the KPI store with version 1.0 and certification badge
- Draft KPIs visible to Analysts only; not surfaced in dashboards or insights until certified

Step 3 - Dashboard Creation

- A sample dashboard is generated best fitting for the selected KPIs and type of project created
- Analyst can also further open Dashboard Studio and select a template or starts from a blank canvas if the sample dashboard is not preferred
- Drags KPI tiles, charts, and widgets onto the canvas; AI recommends chart types per KPI
- Saves as a Team or Executive dashboard and submits for approval
- HITL Workflow Agent routes for sign-off; once approved, dashboard published to the relevant role group

Step 4 - Continuous Monitoring

- Data Agent runs on the configured sync schedule; new data ingested and transformed
- KPI Agent recalculates all certified KPIs and pushes kpi_updated events
- Dashboard tiles update via WebSocket; all views reflect the current period
- Each role sees their filtered view: executive sees strategic summary, manager sees department KPIs

Step 5 - Insight Detection & Delivery

- Insight Agent detects anomaly: e.g., Revenue in Region X down 18% vs 30-day baseline (confidence: 91)
- Explainability Agent constructs the evidence trail: source dataset, KPI formula used, data freshness, detection rationale
- Insight card generated and delivered to the relevant role feeds — executive sees strategic framing, manager sees operational detail

Step 6 - Action & Approval

- Decision Agent classifies: P1 severity, action type = investigation + corrective response required
- Task Assignment Agent identifies Regional Sales Manager as owner based on role mapping

- Approval request sent to Executive via in-app notification and email with insight summary and proposed action
- Executive approves; Task Assignment Agent creates task in the platform and optionally in Jira/Slack/CRM

Step 7 - Execution & Feedback

- Sales Manager receives task with full insight context, recommended action, and due date
- Manager marks task In Progress, executes corrective action, resolves task with a comment
- Manager provides feedback on the insight: accepts it as accurate
- Feedback stored as a positive signal; Insight Agent retains current sensitivity for this KPI pattern
- Resolution triggers an event-driven report dispatched to the Executive and relevant stakeholders

Step 8 - Reporting

- Reporting Agent composes role-specific narratives: executive summary, department performance digest, operational brief
- Scheduled reports delivered at configured frequency; event-triggered reports on P1 resolution
- All reports stored in-app with full version history linked to the KPI snapshot that generated them
- Any report can be sent forward as required on a daily or weekly basis in any form preferred

6. Technical & Architectural Requirements

Layered architecture with clean separation of concerns: UI Layer, AI Agent Core, Data Layer, Security Layer, and Deployment Layer. Two reference views are provided: a component architecture overview and a technical user journey flow.

6.1 Architecture Overview (Component View)

UI Layer

- React SPA (TypeScript) with TailwindCSS and Recharts/D3.js for visualizations
- Components: Dashboard Studio, KPI Builder, Insight Feed, Task Inbox, Report Viewer, Approval Queue, Audit Center, Embedded Copilot
- Role-based view rendering driven by JWT claims; WebSocket for real-time KPI and insight push
- Drag-and-drop dashboard canvas using a grid layout library (e.g., react-grid-layout)

GenAI Core / Agent Orchestration Layer

- Orchestrator: event-driven message broker (Redis Streams or AWS EventBridge) routing typed events between agents
- Nine containerized agent workers (one per agent); each subscribes to specific event types only
- LLM Integration: OpenAI GPT-4o or Anthropic Claude for KPI generation, insight rationale, report composition, and decision reasoning
- Prompts Engine: versioned, templated prompt library per agent use case — stored in DB, hot-reloadable without redeploy
- Structured RAG: pgvector or Pinecone holding business context, KPI definitions, stakeholder mappings, and glossary terms
- HITL Queue Service: manages approval workflows for KPIs, dashboards, and P1 actions with SLA tracking
- Feedback Store: labelled user feedback signals per insight and KPI for iterative agent tuning

Data Layer

- Connector Registry: metadata store for all connected sources, credentials (encrypted), schemas, and sync configs
- Raw ingestion store: S3 or cloud object storage for raw payloads before transformation
- Data transformation: dbt Core models raw data into clean, analysis-ready tables
- Analytical store: Snowflake or PostgreSQL for processed datasets and KPI snapshots
- KPI time-series store: optimized append-only table for trend and anomaly queries
- Vector store: pgvector or Pinecone for RAG context used by KPI, Insight, and Explainability agents

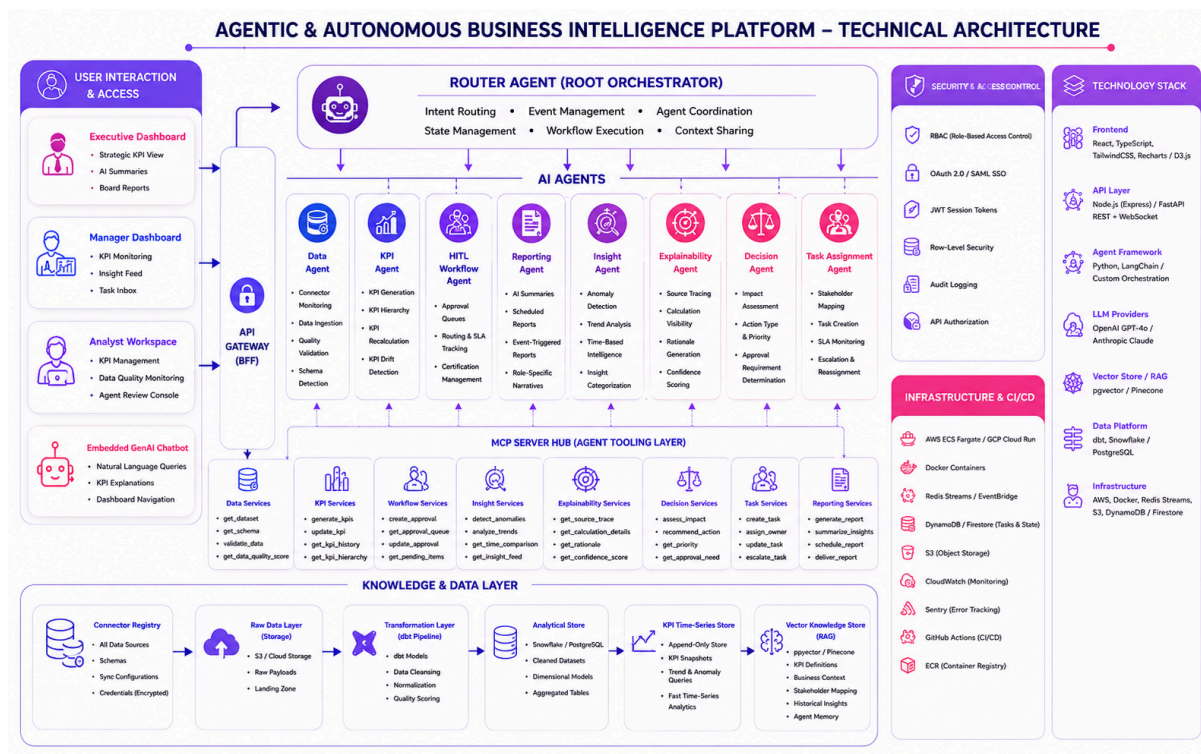
Security Layer

- RBAC enforced at API gateway level on every request; role claims in JWT
- SSO via OAuth 2.0 / SAML; session tokens with refresh token rotation
- Row-level data security applied at the DB query layer — not just API filtering
- Credential encryption for all connector configs using secrets manager (AWS Secrets Manager / GCP Secret Manager)
- Immutable audit log: append-only write; no update or delete operations permitted on log records
- Logical Isolation: Default mode using Tenant_ID filtering at the API and Database (Postgres Row Level Security) levels.

- Physical Isolation (Silo Model): Architecture supports spinning up dedicated database schemas or entirely separate analytical store instances (Snowflake/PostgreSQL) for Tier-1 Enterprise clients.
- Support for Envelope Encryption. Data is encrypted with a Data Encryption Key (DEK), which is wrapped by a Key Encryption Key (KEK) owned by the client (BYOK).

Deployment Layer

- Docker containers orchestrated via AWS ECS Fargate or GCP Cloud Run
- API Gateway as single entry point; rate limiting and request validation at edge
- DynamoDB or Firestore for user state, task records, approval queues, and feedback
- CloudWatch / Cloud Monitoring for observability; Sentry for application error tracking
- CI/CD: GitHub Actions → test → Docker build → registry push → container service deploy
- Async heavy workloads (bulk insight runs, batch reports) handled via AWS Batch or Cloud Run Jobs

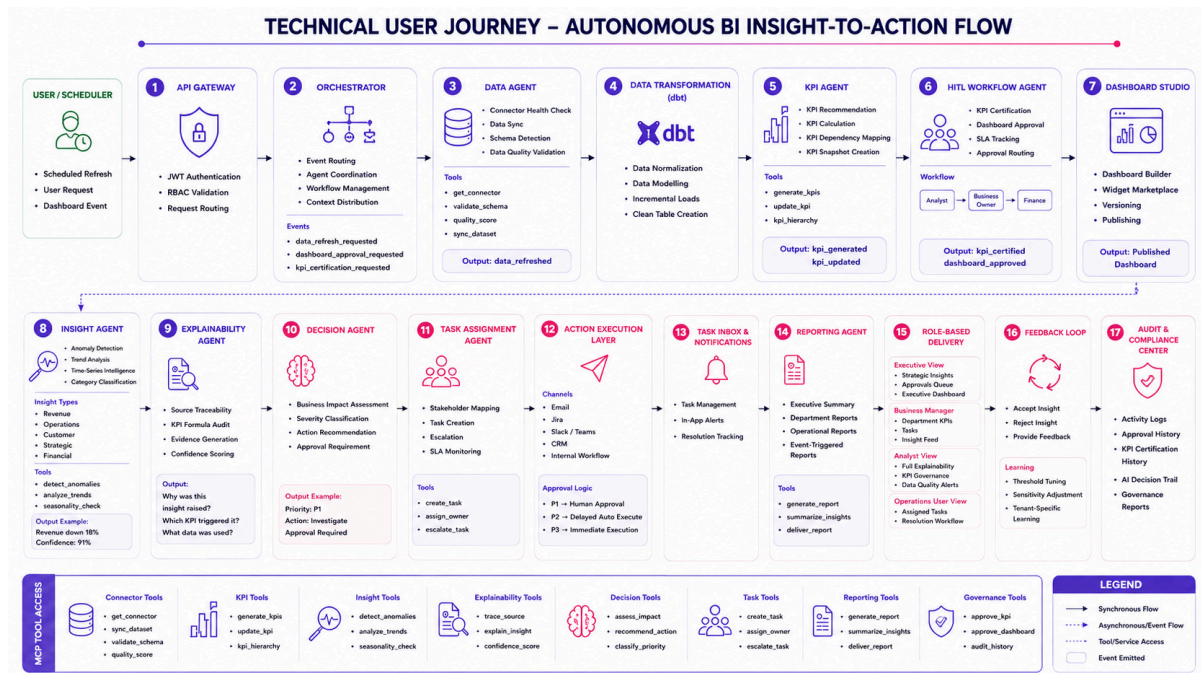


6.2 Technical User Journey Flow

17-step flow mapping a full autonomous cycle from data refresh through action execution across all system components.

- Scheduler or user triggers data refresh via API Gateway
- API Gateway authenticates (JWT + RBAC) and routes to Orchestrator
- Orchestrator publishes `data_refresh_requested` to message broker
- Data Agent pulls from connector, runs quality checks, writes raw payload to object storage
- dbt transformation job runs; clean data written to analytical store
- Data Agent emits `data_refreshed` event with dataset ID and schema fingerprint

7. KPI Agent queries analytical store, recalculates all certified KPIs, stores snapshots in time-series store
8. KPI Agent emits `kpi_updated` events; dashboard WebSocket layer pushes updates to active sessions
9. Insight Agent runs z-score and trend checks against rolling baselines across all certified KPIs
10. Anomaly detected: Insight Agent emits `insight_raised` with category, severity, confidence, and time-period deltas
11. Explainability Agent constructs evidence trail: source trace, KPI formula, data freshness, detection rationale
12. Decision Agent evaluates enriched insight using LLM + RAG context; determines action type, priority level, and approval requirement
13. Task Assignment Agent queries stakeholder role map; creates task record in NoSQL store; routes approval request if P1
14. Notification service dispatches in-app push + email/Slack/Jira depending on configured action channels
15. On approval: action executed, task status updated, resolution notification sent to stakeholders
16. Reporting Agent composes role-specific narrative; stores report; dispatches on schedule or insight resolution trigger



6.3 Technology Stack

Layer	Technology
Frontend	React (TypeScript), TailwindCSS, Recharts / D3.js, react-grid-layout, WebSocket
API Layer	Node.js (Express) or FastAPI (Python) - REST + WebSocket
Agent Framework	Python-based agent workers; LangChain or custom orchestration
LLM Provider	OpenAI GPT-4o or Anthropic Claude API
Vector Store / RAG	pgvector (Postgres extension) or Pinecone

Data Transformation	dbt Core
Analytical Store	Snowflake or PostgreSQL
Message Broker	Redis Streams or AWS EventBridge
Task / State Store	AWS DynamoDB or Firebase Firestore
Object Storage	AWS S3 or GCP Cloud Storage
Secrets Management	AWS Secrets Manager or GCP Secret Manager
Deployment	Docker, AWS ECS Fargate or GCP Cloud Run
Observability	CloudWatch / Cloud Monitoring + Sentry
CI/CD	GitHub Actions
Auth	Auth0 or AWS Cognito (SSO, JWT, RBAC)

7. Project Timeline & Phased Rollout

Two developers. 8–10 weeks. The timeline is structured into four phases with clear deliverables per phase. No phase begins until the prior phase's acceptance criteria are met.

Phase	Focus	Deliverables
Phase 1 (Wk 1-2) Foundation	Infrastructure, auth, first connector	Cloud infra provisioned. CI/CD live. Auth (RBAC + SSO). PostgreSQL connector. dbt pipeline. Dashboard shell.
Phase 2 (Wk 3-5) Core Platform	KPI Engine, HITL, Dashboard Builder, Data Agent	KPI generation + certification workflow. Drag-and-drop dashboard. Role-based views. Real-time KPI updates. Dashboard versioning.
Phase 3 (Wk 6-8) Agent Intelligence	Insight Engine, Action Engine, Explainability, Feedback, Audit	5-category insight detection. Role-filtered insight delivery. Autonomous action pipeline + approval flow. Explainability layer. Feedback controls. Audit center.
Phase 4 (Wk 9-10) Hardening & Launch	QA, performance, UAT, launch	Full pipeline tested end-to-end. Load testing passed. UAT complete. Production deploy. Runbook and handover.

Key Milestones

Milestone	Target Week
Infrastructure & CI/CD live	Week 1
First connector (PostgreSQL) operational	Week 2
KPI generation + HITL certification working	Week 4
Dashboard Studio live with role-based views	Week 5
First insight raised end-to-end	Week 6
Action pipeline + approval flow live	Week 7
Explainability, feedback, and audit center live	Week 8
UAT sign-off	Week 9
Production launch	Week 10

8. Success Metrics (KPIs)

Metrics are split into product performance KPIs (measuring the platform itself) and business impact KPIs (measuring outcomes for users).

8.1 Product Performance KPIs

Metric	Definition	Target
Insight Latency	Time from data refresh to first insight card appearing	< 5 minutes
KPI Certification Rate	% of AI-generated KPIs that pass HITL approval without manual edit	> 70%
Task Assignment Accuracy	% of tasks routed to the correct stakeholder (verified in UAT)	> 90%
Anomaly Precision	% of raised insights confirmed as genuine by users	> 80%
Explainability Coverage	% of insights with full evidence trail attached	100%
Action Execution Rate	% of approved actions successfully executed across channels	> 99%
Dashboard Load Time	P95 time for dashboard initial render with full KPI data	< 3 seconds
Agent Pipeline Uptime	% uptime for the full agent orchestration pipeline	> 99.5%
Feedback Loop Adoption	% of insight cards that receive Accept/Reject/Feedback signal	> 50% within 30 days of launch

9. Assumptions, Risks & Dependencies

9.1 Assumptions

- Data sources provide structured access via supported connectors — no unstructured or OCR-dependent data in MVP
- LLM API remains available and within latency/cost budget for MVP workloads
- Business has stakeholder-to-role mappings available to seed the Task Assignment Agent
- At least one business stakeholder is available to participate in HITL KPI certification during Phase 2
- Two developers are full-stack capable and comfortable with AI/agent infrastructure and cloud deployment
- Cloud infrastructure costs are pre-approved for the development and launch phase
- This is a greenfield deployment — no legacy system migration in scope
- The system is designed as Multi-Tenant, requiring a `tenant_id` on every database record and API request.
- Agent memory (Vector Store context) is partitioned per tenant; no cross-tenant data retrieval is possible via the RAG (Retrieval-Augmented Generation) layer.
- The platform's isolation strategy is designed to meet SOC2 and GDPR requirements regarding data residency and portability.

9.2 Risks & Mitigations

Risk	Impact	Mitigation
LLM quality insufficient for KPI generation on complex schemas	High - core feature fails	HITL approval gate catches poor suggestions before publish; manual KPI builder as fallback
HITL bottleneck: business approvers slow to certify KPIs	High - dashboard launch blocked	Configurable auto-approval after SLA timeout; draft KPI preview for analysts while awaiting certification
Data quality of connected sources is poor	High - insight accuracy degrades	Data Agent quality scorecard; low-quality datasets quarantined before agent pipeline
False positive insight rate is high	Medium - user trust erodes	Feedback loop enables per-KPI sensitivity tuning; confidence score threshold configurable per role
Agent pipeline latency too high	Medium - UX degrades	Async execution with optimistic UI updates; WebSocket push avoids polling
Action channel integrations (Jira, Slack) add scope	Medium - timeline risk	Each integration is a plugin; deliver Email channel first, add others incrementally
Scope creep with 2 developers	High - timeline slips	Strict phase gates; nothing added post-Phase 2 kickoff without equivalent removal
LLM API costs exceed budget	Medium - commercial risk	Prompt caching, batching non-real-time agent calls, model routing to cheaper models for low-stakes tasks

Security gap in multi-tenant data access	High - data breach	Row-level security at DB layer; secrets manager for credentials; penetration test pre-launch
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9.3 Dependencies

Dependency	Owner / Risk
LLM API access (OpenAI / Anthropic)	External - provisioned before Phase 2
Cloud account and IAM setup	DevOps - complete by Week 1
Initial dataset from business stakeholder	Business team - needed by Week 2
Business approvers for HITL KPI certification	Business / Finance - engaged by Week 3
Stakeholder role mapping document	Business / HR - needed by Week 6
Auth provider setup (Auth0 / Cognito)	DevOps - end of Phase 1
Action channel credentials (Jira, Slack, CRM)	IT / DevOps - needed by Phase 3
UAT participants from target user personas	Product team - confirmed by Week 8